CompoundIQ — Executive Overview

"Expedia for Compounding Pharmacies"

What Is CompoundIQ?

CompoundIQ is a platform that connects functional medicine clinics with compounding pharmacies. It handles everything from finding the right pharmacy to collecting patient payment to tracking the prescription through delivery — replacing what is currently a manual, phone-and-fax-driven process.

Think of it as the infrastructure layer that sits between the clinic and the pharmacy, making the entire ordering process faster, more transparent, and more profitable for everyone involved.

The Problem We Solve

Functional medicine clinics prescribe compounded medications every day, but the process of actually getting those prescriptions filled is broken:

Finding the Right Pharmacy Is a Guessing Game

When a patient needs a compounded medication, the medical assistant has to manually research which pharmacies carry the right formulation, are licensed in the patient's state, offer competitive pricing, and can deliver quickly. This happens through phone calls, PDF price lists, and personal relationships. There is no searchable database, no comparison tool, and no way to verify state licensing in real time.

What CompoundIQ does: Our sourcing engine lets the medical assistant search all pharmacies instantly, filtered by the patient's state. Only pharmacies with active licenses in that state appear in results — illegal out-of-state dispensing is prevented automatically at the system level, not by human memory.

Clinics Leave Money on the Table

Most clinics mark up compounded medications (a legal and common practice in functional medicine), but they calculate margins manually. The medical assistant looks up a wholesale cost, picks a retail price, and does the math on paper or in their head. There are no tools to visualize margin in real time, no safeguards against selling below cost, and no transparency into platform fees.

What CompoundIQ does: Our margin builder shows the wholesale cost (locked and read-only), lets the clinic set their retail price with one-click multiplier buttons (1.5x, 2x, 2.5x, 3x), and instantly calculates the margin percentage, platform fee, and estimated clinic margin. A database-level constraint prevents selling below cost. The clinic sees exactly what they earn before committing.

Payment Collection Is Disconnected from the Prescription

Patients typically pay by calling the clinic with a credit card number, paying at the front desk, or being told to pay the pharmacy directly. None of these methods are connected to the prescription. This creates delays, cart abandonment with no follow-up, and compliance risk from non-PCI-compliant payment handling.

What CompoundIQ does: After the provider signs the prescription, the system generates a secure payment link and sends it to the patient via text message. The patient taps the link on their phone,

sees a clean checkout page branded with their clinic's name, enters their card (or uses Apple Pay / Google Pay), and pays. No login, no app download, no account creation. The payment automatically splits three ways — the clinic gets their payout, the platform captures its fee, and it all happens in a single transaction.

Fulfillment Is a Black Box

Today, prescriptions are faxed to the pharmacy. Then everyone waits. There is no confirmation the fax was received, no tracking of when compounding starts, no automated alerts when things stall, and no visibility for the patient. If something goes wrong, the clinic finds out when the patient calls to ask where their medication is.

What CompoundIQ does: Our system routes prescriptions through the best available channel for each pharmacy. Some pharmacies have modern APIs and get instant digital delivery. Others have web portals that our system fills out automatically. The rest get fax — but even fax is tracked, with automatic retries and escalation if delivery fails. Every step is monitored by our SLA engine, which automatically alerts the operations team when deadlines are missed.

How It Works — The User Experience

For the Medical Assistant (30 seconds per prescription)

1. **Select patient & provider** — Search for the patient by name. The provider auto-selects if the clinic has only one. Both stay pinned at the top of every screen via a persistent session banner.
2. **Configure medication** — Search by ingredient name. The system reveals options progressively: salt form (Cypionate vs Enanthate), formulation (concentration, oil base), then dose, frequency, timing, and duration. The structured sig builder auto-generates prescription directions with unit conversion (mg, mL, and syringe units for injectables). For titration protocols (like LDN), toggle Titration mode and enter start dose, increment, interval, and target — the sig generates automatically. For cycling protocols (like Thymosin Alpha-1), toggle Cycling mode with on/off day patterns.
3. **Or use shortcuts** — Load a saved Provider Favorite in one click (pre-fills the entire prescription), or apply a Clinic Protocol Template to add multiple medications to the session at once (e.g., "Weight Loss Protocol" adds Semaglutide + BPC-157 + Lipo-Mino in one click).
4. **Price** — Select a pharmacy (filtered by patient's state), set the markup with one click. See the margin instantly.
5. **Add more or send** — "Add & Search Another" to add more prescriptions for the same patient, or "Review & Send" to batch review everything. For a 3-medication visit, this takes about 45 seconds total.
6. **Or save as draft** — If the provider isn't available, the MA saves the prescription as a draft. The provider logs in later, reviews from the Drafts tab, and signs with one tap.
7. **Review & sign** — The batch review page shows all prescriptions with drug interaction alerts if any known conflicts exist. The provider signs once — a single signature covers all prescriptions in the session. If any prescription contains a controlled substance (DEA Schedule II-V), EPCS two-factor authentication is required at the point of signing (TOTP authenticator app on a separate device, per DEA 21 CFR 1311).
8. **Done** — Payment links fire automatically. SMS reminders at 24 and 48 hours if the patient hasn't paid. Auto-cancellation at 72 hours. No follow-up needed.

For the Patient (60 seconds)

1. **Receive a text** — "Hi Alex, Dr. Chen at Sunrise Functional Medicine has finalized your custom prescription. Tap here to pay."
2. **Tap the link** — Opens in their phone's browser. No app, no login.
3. **See the checkout** — Clinic name and logo. "Prescription Service — $300.00." No medication details shown (privacy).
4. **Pay** — Apple Pay, Google Pay, or card. One tap.
5. **Confirmation** — Animated checkmark, order reference, "Within 3–7 business days" or "Within 24–48 hours" depending on the pharmacy.

For the Operations Team (Continuous Monitoring)

1. **Pipeline view** — See every order across every clinic in one dark-mode dashboard.
2. **SLA tracking** — Color-coded deadlines. Red means overdue. Automatic Slack alerts and escalation.
3. **Pharmacy health** — Monitor which pharmacy integrations are healthy, degraded, or down.
4. **Fax triage** — When pharmacies respond via fax, the ops team matches responses to orders.
5. **Catalog management** — Upload and manage pharmacy medication catalogs with version tracking.

The Pharmacy Adapter — Our Core Innovation

The compounding pharmacy landscape is fragmented. Some pharmacies have modern technology (APIs), some have web portals, and many still rely entirely on fax. No existing platform connects to all of them.

CompoundIQ's Pharmacy Adapter Layer solves this with a four-tier integration strategy:

Tier 1 — Direct API Connection For pharmacies with modern REST APIs. The prescription is delivered digitally in seconds, and the pharmacy sends real-time status updates back to the platform. This is the fastest and most reliable method.

Pharmacies: ReviveRX, Vios, MediVera, LifeFile network (Empower, Belmar, UCP, Strive), Precision

Tier 2 — Automated Portal Submission For pharmacies that have web portals but no API. Our system uses a headless browser to log into the pharmacy's portal, fill out the order form, and submit it — automatically. An AI confidence engine verifies the submission was successful by analyzing a screenshot of the confirmation page.

Pharmacies: Olympia/DrScript, Wells/WellsPx3

Tier 3 — Standardized API Specification A published API specification that pharmacies can adopt to integrate with CompoundIQ. Same benefits as Tier 1 but initiated by the pharmacy. This is the long-term growth strategy — once published, it enables rapid partner onboarding.

Pharmacies: Future partners

Tier 4 — Fax Fallback The universal fallback. The system generates a compliant prescription PDF and faxes it via HIPAA-compliant fax service. Delivery is tracked, retries are automatic, and pharmacy responses are triaged by the ops team.

Pharmacies: Any pharmacy not on Tier 1, 2, or 3

The key insight: The system always uses the highest available tier for each pharmacy. If a higher tier fails, it automatically falls down to the next one. Fax is always available as the safety net. The clinic doesn't need to know or care which tier is used — the system handles it transparently.

The LifeFile advantage: LifeFile is the de facto B2B standard for compounding pharmacies. A single LifeFile API integration unlocks an entire network of major pharmacies — Empower, Belmar, UCP, and Strive. This is the single highest-leverage integration in the space.

Revenue Model

CompoundIQ earns revenue on every transaction through a transparent platform fee:

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Patient pays:          $300.00 (retail price set by clinic)
Pharmacy receives:     $150.00 (wholesale cost)
Clinic earns:          $127.50 (their markup minus platform fee)
Platform fee (15%):     $22.50 (15% of the $150 margin spread)
```

- The platform fee is a percentage of the **margin spread** (retail minus wholesale), not the total transaction
- The clinic sees the exact fee before they commit — full transparency
- Stripe processing fees (2.9% + $0.30) come out of the platform's portion, not the clinic's
- No subscription fees, no per-user fees — purely transaction-based

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Security & Privacy

CompoundIQ handles protected health information (PHI) and is designed for HIPAA-adjacent compliance:

Patient privacy is enforced by the system, not by trust:

- The patient checkout page never shows medication names — only "Prescription Service"
- Stripe (our payment processor) never sees any health information — only an order ID
- Text messages to patients contain only their first name and a payment link — never medication details
- Each clinic can only see their own patients and orders — enforced at the database level, not just the application

All credentials and sensitive data are encrypted:

- Pharmacy API keys and portal passwords are stored in an encrypted vault, never in plain text
- All data is encrypted at rest (AES-256) and in transit (TLS 1.2+)
- Patient payment information never touches our servers — it goes directly to Stripe

Access is strictly controlled:

- Medical assistants and providers can only access their own clinic's data
- The operations team has cross-clinic visibility but cannot modify clinical records
- Patients can only see their own order through a time-limited, cryptographically signed URL
- Sessions automatically expire after 30 minutes of inactivity

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The Competitive Landscape

Platform	What They Do	What They Don't Do
Rupa Health	Lab test ordering for functional medicine	No pharmacy fulfillment, no compounding

PrescribeWellness	Retail pharmacy adherence	No compounding, no multi-pharmacy sourcing
Pharmacy Portals (DrScript, WellsPx3, LifeFile)	Single-pharmacy ordering	No cross-pharmacy comparison, no payment, no unified tracking
E-Fax Services	Send faxes	No intelligence, no tracking, no escalation, no payment
CompoundIQ	End-to-end: search, price, pay, route, track, deliver	—

No platform today provides: cross-pharmacy search with state licensing compliance + cascading dropdown prescription builder with structured sig generation + titration and cycling protocol support + provider favorites and clinic protocol templates + EPCS two-factor authentication for controlled substances + drug interaction alerts + integrated margin calculation + patient SMS checkout with multi-party payment routing + intelligent prescription routing across 4 submission methods + unified order lifecycle tracking with SLA enforcement.

Current Status

Milestone	Status
Platform built	19 phases, 87 work orders — all merged to main
All 3 applications functional	Clinic App, Ops Dashboard, Patient Checkout
Cascading prescription builder	Hierarchical catalog, structured sig, titration/cycling
Provider speed features	Favorites (one-click reorder), protocol templates (multi-med bundles)
Regulatory compliance	EPCS 2FA for controlled substances, drug interaction alerts
Automated test coverage	65 Playwright E2E tests across Chromium / Firefox / WebKit / mobile Chrome, plus 6 jest unit tests. CI gates every merge to main — `continue-on-error` removed.
Externally tested	Claude Cowork browser QA validated all phases + the full 45-minute demo walkthrough across 6 rounds with zero remaining findings
Documentation	60+ specification documents organized in software factory
POC deployed	Live on Vercel, accessible for demonstration
Security audit	AES-256 encrypted TOTP secrets, RLS on all 33 tables

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What's Next

Near-term:

- Onboard 2–3 design partner clinics for beta testing
- Activate Twilio SMS for real patient notifications
- Activate Documo fax for real pharmacy submissions
- Complete Stripe Connect onboarding for first clinic

Medium-term:

- Integrate LifeFile API (unlocks Empower, Belmar, UCP, Strive)
- Publish Tier 3 API specification for pharmacy self-onboarding
- Launch Tier 2 portal automation for Olympia and Wells

Long-term:

- Scale to 50+ pharmacy partners across all 4 tiers
- Expand to additional therapeutic categories beyond functional medicine
- Build analytics and reporting for clinic revenue optimization

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Key Talking Points

For investors:

> "*CompoundIQ is the Expedia for compounding pharmacies — a B2B2C infrastructure layer that captures a 15% fee on every prescription transaction. The 4-tier adapter strategy means we can onboard any pharmacy regardless of their technology, and the LifeFile integration alone unlocks the largest compounding pharmacy network in the country. Our cascading prescription builder with titration/cycling protocols, provider favorites, and clinic protocol templates is a market-first — no competitor has structured sig generation or one-click multi-medication bundles.*"

For clinic partners:

> "*Your medical assistant configures a prescription in 30 seconds with cascading dropdowns — ingredient, form, dose, frequency, done. Save it as a favorite for one-click reorder. Build protocol templates that add 3-5 medications in a single click. The provider signs once for the whole batch. The patient pays from their phone. You earn margin on every fill with full transparency. And you never touch a fax machine again.*"

For pharmacy partners:

> "*You receive structured, error-free orders instead of handwritten faxes. Real-time status updates flow back to the clinic automatically. And our standardized API spec means you can integrate once and receive orders from every clinic on the platform.*"

For patients:

> "*You get a text with a payment link. Tap, pay, done. Your prescription is on its way. You'll get updates by text as it progresses.*"

CompoundIQ — Intelligent sourcing. Automated fulfillment. Transparent pricing.